

Lab 1: Requirements Table

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Problem Statement: #28 - Inter-City Freight Load Matching Marketplace

Scenario: A B2B freight brokerage exchange where commercial shippers post cargo loads and trucking carriers submit bids, with automated milestone payment release upon proof-of-delivery sign-off.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.	High	Pass: a bid submitted by a verified carrier during the specified auction timeframe is recorded and the shipper is notified. Fail: an unverified carrier is able to submit a bid, or a bid submitted after the auction timeframe is accepted.	Bidding is the core marketplace interaction between shippers and carriers.	
FR-002	Functional	The system shall allow a shipper to post a freight load by entering origin, destination, cargo type, weight, and pickup and delivery time windows.	High	Pass: after all required fields are submitted, the load is published and visible to verified carriers. Fail: a load with any required field missing is published.	Carriers need complete shipment details before deciding whether to bid.	
FR-003	Functional	The system shall allow a shipper to review submitted bids and award a load to exactly one carrier before the auction deadline. If no bid is awarded, the system shall automatically close the auction as unawarded at the deadline.	High	Pass: selecting 'Award Bid' before the deadline locks the auction, notifies the winning and losing carriers, and sets the load status to 'Awarded'. If no bid is selected, the auction closes automatically at the deadline as 'Closed - Unawarded'. Fail: more than one carrier is awarded the same load, or an unawarded auction remains open after its deadline.	Each shipment needs one responsible carrier, and every auction must reach a clear final state.	Peer review: clarified the valid unawarded outcome and automatic deadline closure.
FR-004	Functional	The system shall allow the awarded carrier to submit an electronic proof of delivery (e-POD) containing a digital signature, timestamp, and GPS geo-tag after completing delivery.	High	Pass: the awarded carrier submits an e-POD containing all three required components; it is recorded and the shipment status changes to 'Delivered - Pending Verification'. Fail: a non-awarded carrier can submit the e-POD, or an e-POD missing any required component is accepted.	A complete e-POD from the awarded carrier is required before delivery can be verified and payment can be released.	

Lab 1: Requirements Table (continued)

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-005	Functional	The system shall allow shippers and carriers to view the current status of any load they are associated with, including 'Posted', 'Awarded', 'In Transit', 'Delivered - Pending Verification', 'Delivered - Payment Released', 'Payment Release Failed - Manual Review', and 'Closed - Unawarded'.	Medium	Pass: an associated shipper or carrier can view the load status, and the displayed status matches the backend state within 15 seconds of a status change. Fail: an unrelated user can view the load, or the displayed status is more than 15 seconds out of date.	Status visibility reduces uncertainty while restricting shipment information to involved parties.	
NFR-001	Nonfunctional	The system shall automatically disburse milestone payments held in escrow to the carrier within 60 seconds of a verified electronic proof-of-delivery submission, and payment-release communication with the Payment Gateway shall be encrypted.	High	Pass: under simulated peak load, a verified e-POD results in payment disbursement within 60 seconds of its submission and Payment Gateway communication is encrypted. Fail: payment takes longer than 60 seconds, or Payment Gateway communication is not encrypted.	Fast, secure payout is important to carrier trust and protects payment data.	
NFR-002	Nonfunctional	The system shall maintain at least 99.9% availability for load posting, bidding, and shipment tracking over each calendar month.	Medium	Pass: monthly availability logs show at least 99.9% uptime for load posting, bidding, and shipment tracking. Fail: monthly availability for any of those services is below 99.9%.	High availability helps prevent missed bids and interruptions to active shipments.	